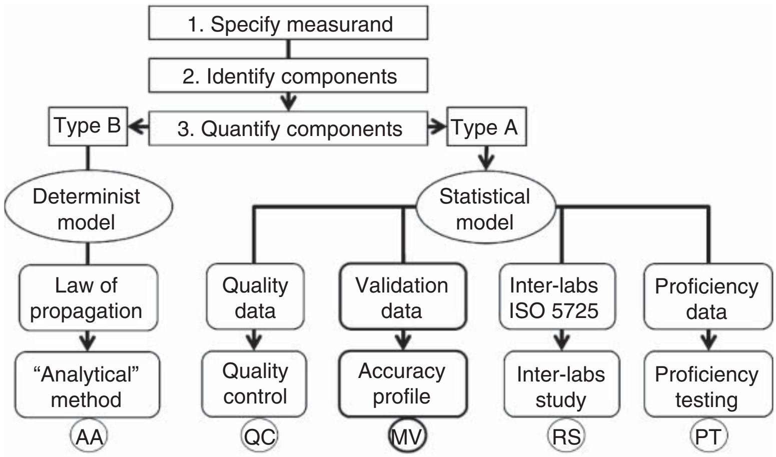

Figure 7.1 Different MU estimation procedures proposed for the analytical sciences. The two-letter codes refer to the list of documents in Table 7.1.

Table 7.1 Selected references of consensus methods for estimating measurement uncertainty in analytical sciences.

Code	Organization	References	Years
AA	BIPM – GUM	[2]	2012
AA	EURACHEM-CITAC	[3]	2012
MV	SFSTP	[4]	2017
MV	AFNOR	[5]	2010
MV	EDQM-OMCL	[6, 7]	2020
MV	ISO	[8]	2012
PT	ISO	[9]	2015
PT	EDQM-OMCL	[10]	2020
QC	EDQM-OMCL	[11]	2020
QC	ISO	[8]	2012
RS	ISO	[12]	2010
RS	EDQM-OMCL	[13]	2020

The organization acronyms are explained in Section 12.3.

– The direct consequence is that a harmonization effort promoted by analysts is crucial to propose the most judicious procedure or procedures, even if several approaches have already been the subject of recommendation or regulation.

Moreover, some specialized scientific publications describe dedicated applications. The journal *Accreditation and Quality Assurance* has published a few hundred examples for the analytical sciences. Even the vocabulary can be affected by these discrepancies in some fields of application, such as medicines control, where two approaches are distinguished:

- *Top-down* to characterize the Type A evaluation based on data collected during interlaboratory or inter-comparison studies (designated by the code RS).
- *Bottom-up* is often based on a deterministic model and is synonymous with Type B evaluation.

The distinction between bottom-up and top-down does not shed much light on eventual harmonization [14]. Nevertheless some publications conclude that obtained values are nearly equivalent [15]. Several meaningful and suitable approaches are presented in the remainder of this chapter. They were selected because they allow reusing data that has already been collected for another purpose and do not require any new experimental effort. In that respect, the data already used for the method accuracy profile (MAP) is particularly appropriate. The main reason to recommend this procedure is that data collection follows strict rules fully compliant with MU estimation.

7.2 Use Method Accuracy Profile Data

7.2.1 Stage 1. Generic Measurement Model

To illustrate why we consider that MAP procedure can be an attractive approach to estimate MU, the different steps of general GUM procedure presented in Section 6.2 are inspected in detail, keeping in mind this specific application. Some publications present this extension of the accuracy profile to estimate MU, for example for a statin analytical method [16]. The first step is to formalize a measurement model that applies to all methods and avoids overly specific applications. The ISO/TS 21748:2017 standard can serve as a guide. Its scope covers two topics:

- MU evaluation is based on data obtained from interlaboratory studies organized in accordance with ISO 5725.
- The comparison of the MU, calculated from an interlaboratory study, with that obtained by applying the principles of the law of propagation of uncertainty.

In this standard, the proposed measurement model is generic and applicable to any field of measurement. According to the GUM recommendations, the calculation procedure can combine the Type A and B approaches. Before going any further, it should be remembered that MU characterizes a measurement made in a single laboratory and not the method applied in different laboratories, as is the case with reproducibility. It would therefore be nonsense to compare measurement uncertainty with a standard deviation of reproducibility [17]. As it is inconsistent to use the results of an interlaboratory analysis to estimate individual MU, several precautions are taken in ISO/TS 21748:2017 standard to avoid this problem.

When data are collected in a laboratory under intermediate precision conditions, there is no longer any contradiction in using the measurement model developed in this standard, even though it was developed for interlaboratory comparisons. To obtain a MAP, it is mandatory to collect data under intermediate precision conditions, as explained in Section 5.4.2.

The similarity between the computation of reproducibility and intermediate precision variances was underlined in Section 3.2.1. In addition, since a full concentration range is covered with the accuracy profile, it will be possible to obtain MU values over the validation range at all concentration levels. Finally, an uncertainty function to calculate the MU for any concentration could be obtained. This function will be helpful in calculating the MU of any unknown sample result.

The ISO 21748:2017 measurement model principally considers a major part of the analytical process, i.e. the analytical operating procedure, and includes some pre-analytical steps. This applies to all analytical methods. It is presented in the form of the Eq. (7.1).

Generic measurement model

$$Z = X + \delta + B + \sum_{p=1}^{P} c_p G_p + E \tag{7.1}$$

where:

- Z Recovered (or inverse-predicted) concentration of the analyzed material.
- X Reference value assigned to the analyzed material (it is a constant).
- δ Intrinsic bias of the measurement method.
- B Random variation is produced by the series factor effect that combines many sources of uncertainty.
- G_p Other input quantities introduce a deviation from the nominal value of the reference X.
- c_p Sensitivity coefficient of the input quantity (see Eq. 6.18).
- E Residual random error, under repeatability condition.

The quantity $\sum_{p=1}^{P} c_p G_p$ represents the part of the measurement model that considers the slight deviations between the assigned value and the observed measurement values, e.g. during the preparation of the calibrators. As before, the sensitivity coefficient c_p reflects the relative contribution of the quantity G_p in the MU.

7.2.2 Stage 2. Generic Cause-to-Effect Diagram

In practice, the recommended starting point for listing MU components is to draw a cause-to-effect diagram, as described in Section 6.5. If approach B is applied, this is restricted to the input quantities involved in the formula applied for result expression. This limitation has been criticized, as it appears to ignore many sources of uncertainty that play a fundamental role in the measurements, such as sample preparation procedures or instrument settings. Table 7.2 presents a brief list of sources conventionally described in the literature in relation to sampling or sample preparation.

In the ISO 17025:2017 standard, there is an explicit list of critical points to be controlled in a laboratory to reduce the "variability," and the resulting uncertainty.

Table 7.2 Sources of uncertainty in sampling and sample preparation.

Sampling	Preparation of the sample
Heterogeneity (or inhomogeneity)	Homogenization and subsampling
Influence of the sampling strategy (random, stratified, proportional, etc.)	Drying
	Grinding
Sedimentation effects of bulk material	Dissolution
	Extraction
Bulk physical state (solid, liquid, gas)	Contamination
Effects of temperature and pressure	Derivatization (chemical effects)
Influence of sampling on the composition	Dilution errors
	Preconcentration and concentration
Transport and storage of the sample	Effects due to speciation

Source: Adapted from EURACHEM-EUROLAB [18].

Through the reading of the different chapters, a list of practical requirements can be sorted out. It is the role of the inspector sent by the certification body, such as the French accreditation committee Comité français d'accréditation (COFRAC) or the American Association for Laboratory Accreditation (A2LA), to review this list and decide whether they are correctly implemented and controlled in the audited laboratory. Two main types of requirements can be identified corresponding to:

– The "commitment of results" where the laboratory is invited to participate in proficiency testing and demonstrate that it is capable of achieving acceptable results or scores.
– The "commitment of means" (or resources) where the laboratory must mobilize all human and technical aspects to ensure the performance, prudence, and diligence of its service at the declared level of quality.

These obligations give an insight into what must be considered as critical points for controlling the sources of MU. Several chapters of ISO 17025:2017 can be seen as participating in the catalog of MU components. It is adequate to organize them into a *generic* cause-to-effect diagram applicable to any analytical method issued from any laboratory, accredited or not.

Figure 7.2 represents a proposal summary to illustrate such a diagram. It differs from Figure 6.2 in that it identifies eight sources of uncertainty corresponding to eight chapters of ISO 17025:2017 [1]. It sometimes takes its name, Ishikawa diagram, from the Japanese quality promoter who proposed it in 1943. In an industrial context, the causes are categorized by the "5 M's" for machine, method, material, man/mind power, and measurement/medium. This diagram can also be called "5 M's".

As recommended, several sources of uncertainty can be clustered as one dotted area labeled *series effect* on the figure. This also appears in the measurement model of Eq. (7.1) as the B random variable affecting the impact of series changes. It is intended to account for a majority of uncertainty sources that are present during the

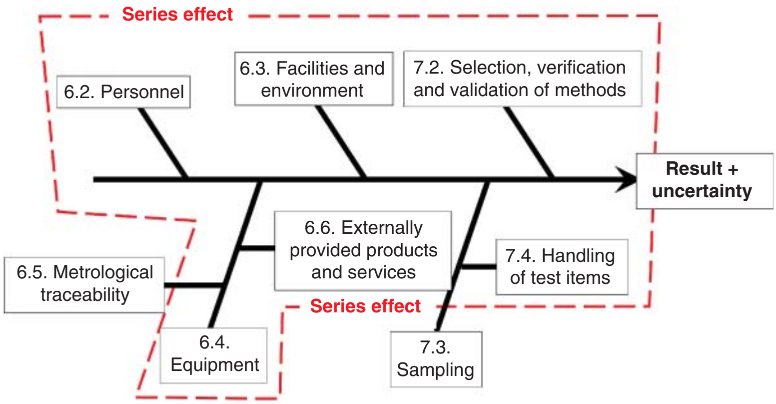

Figure 7.2 Generic cause-to-effect diagram with eight main classic sources of uncertainty, according to [1]. The numbers in each box refer to the chapters of the standard. The dotted area groups sources into one series factor effect.

analytical part of the measurement process. To identify the standard uncertainty of this major part of MU, it is noted $u(Z_m)$ in the following, where subscript m stands for method. This observation ignores two other components of MU:

- $u(Z_s)$ the sampling standard uncertainty, which mainly characterizes the pre-analytical step and is discussed in Section 8.3.
- $u(Z_d)$ the so-called definitional uncertainty, which corresponds component of measurement uncertainty "resulting from the finite amount of detail in the definition of the measurand."

The comprehensive generic measurement model then takes the following form:
Comprehensive generic measurement model

$$Z = \left[\underbrace{X + \delta + B + \sum_{p=1}^{P} c_p G_p + E}_{\text{Analytical process}} \right] + S + D \tag{7.2}$$

where S and D are random variables representing the random errors due to the sampling and the definition of the reference value X, respectively. After applying the law of propagation of uncertainty, the combined standard uncertainty of Z thus becomes:

Combined uncertainty of Z

$$u_c(Z) = \sqrt{u^2(Z_m) + u^2(Z_s) + u^2(Z_d)} \tag{7.3}$$

In the first instance, the two standard uncertainties $u(Z_s)$ and $u(Z_d)$ are neglected to focus only on the standard uncertainty related to the method. This is realistic because, for many laboratories, the sampling (addressed in Section 8.3) is not part of their mission. As explained in Section 6.3 about the traceability to the SI units, the uncertainty on the reference value is negligible compared to the other sources.